EPS QKD

Entanglement-based Quantum Key Distribution Experimentation Kit for Academia

Quantum Key Distribution Experimentation Kit using Entangled Photon Source for academic exploration and benchmarking of entanglement based experiments.

Product Description

The Entangled Photon based Quantum Key Distribution Experimentation Kit features an Entangled Photon Source (EPS) designed with orthogonally placed Paired-BBO crystal in Type 1 Non-collinear Configuration which produces high brightness entanglement pairs with controllable stability. Pumped by a Continuous Wave source at 405nm, it generates down converted photons by Spontaneous Parametric Down Conversion (SPDC) at 810nm. The Signal and Idler are separated by high reflectivity mirror pairs and are coupled into fiber couplers integrated with collection optics.

Building upon the technical analysis of the performance and quality of entangled photons generated, the Kit enables experimentation of Quantum Key Distribution setup with BBM92 protocol. This QKD platform facilitates realising Key Distribution, Quantum Bit Error Rate calculations along with analysis of signal-to-noise ratio all in one inclusive unit. The product is designed as an Integrated experimental QKD system facilitating study of quantum entanglement phenomena and its application for Quantum Key Distribution.

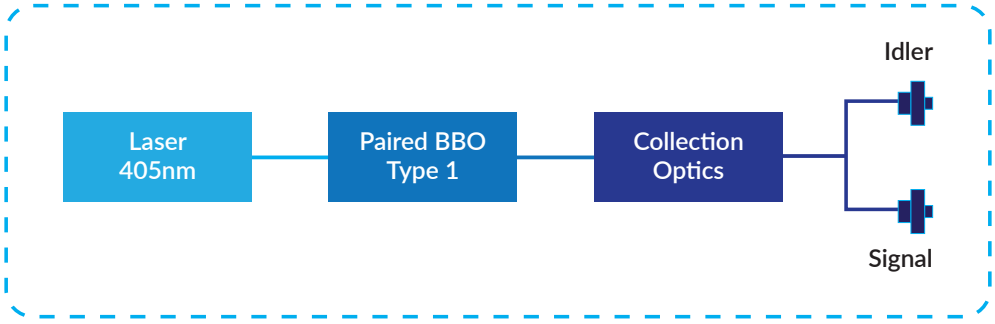


Figure 1: Schematic of EPS in Type 1 configuration for Experimentation Kit



EPS-QKD Experimentation Kit

Product Specifications

Parameter	Value
Entanglement Mode	Polarization
Crystal	Paired BBO Type-1
Emission Wavelength	810nm
Pair Generation Rate	~50kcps
Fidelity	> 90%
Heralding Efficiency	~ 0.08-0.2
Coincidence Counts	~5%
Second Order Correlation Function	< 0.1
|S| - Value	> 2.4
Visibility	> 95%
Thermal Operation Range	10-30 °C
Laser Pump Wavelength	405nm
Laser Pump O/P power	up to 100 mW
Dimensions	550*550*100 mm^3
Weight	15kg

Key Features

- Compact and Portable Entanglement Generation with Non-linear SPDC process.
- Entanglement study with Quantum State Visibility, Quantum State Tomography, Bell Inequality, etc.
- High source brightness with High Fidelity.
- Experimentation HBT experiment, Heralded Single Photon Source.
- Implement BBM92 Entanglement based QKD protocol and more.
- QBER and Secure Key Rate calculations.

Experimentation Outcomes

- Quantum Entanglement with Non-linear optical processes, namely SPDC, facilitated by BBO crystal pair.
- Entanglement source characterization by Single Channel Count Rate, Coincidence Rate, Quantum State Visibility, Heralding Efficiency, Quantum State Fidelity, Quantum State Tomography, Second-order Correlation Function, etc.
- Analyse channel efficiency by calculating Detection Error Rate, QBER, S/N ratio, Secure Key Rate.
- Entanglement mode characterization and coupling efficiency with varying range of Optical Fibers.



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